

ADVANCES IN HEART FAILURE, MECHANICAL CIRCULATORY SUPPORT AND TRANSPLANT

Aortic Stenosis and Heart Failure With Preserved Ejection Fraction: A Cautionary Tale of Common Partners

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ABSTRACT: Aortic stenosis (AS) is increasingly recognized as a disease of both the valve and the myocardium. Beyond valvular obstruction, many patients with aortic stenosis develop extra-valvular abnormalities, including left ventricular hypertrophy, diastolic dysfunction, and atrial and pulmonary vascular remodeling, which have emerged as major determinants of symptoms and prognosis. These abnormalities share many similarities with heart failure with preserved ejection fraction (HFpEF), a condition that is also prevalent in the same aging population. Emerging data suggest that AS and HFpEF frequently coexist and interact, compounded by shared cardiometabolic risk factors, producing convergent phenotypes that complicate diagnosis and influence treatment response. Residual HFpEF appears to underlie much of the persistent heart failure burden after aortic valve replacement, while even mild AS portends worse outcomes in patients with HFpEF. Recent studies have shown that HFpEF-directed therapies, such as sodium-glucose cotransporter 2 inhibitors, may benefit selected patients with AS. As both conditions increase in prevalence and valve interventions are offered to increasingly complex patients, a clearer understanding of the ways in which AS and HFpEF overlap and interact is essential. This review integrates epidemiological, pathophysiological, and clinical perspectives to synthesize emerging evidence on the AS-HFpEF overlap and outlines implications for diagnosis, prognosis, and management.

Key Words: aging ■ glucose ■ heart failure ■ myocardium ■ phenotype

Aortic stenosis (AS) and heart failure with preserved ejection fraction (HFpEF) are among the most common cardiovascular conditions affecting older adults.^{1,2} Traditionally considered distinct entities, they have largely been studied in isolation, with limited investigation of their interaction. However, emerging data suggest that AS and HFpEF are more intricately linked than previously recognized. Both are increasing in prevalence with population aging and share common cardiometabolic risk factors.^{1,2} Furthermore, the cardiac remodeling that accompanies AS involves structural, functional, and cellular changes that closely resemble the hallmark features of HFpEF,³ and incomplete resolution of these abnormalities is associated with persistent heart failure

(HF) and poor outcomes after aortic valve replacement (AVR).^{4,5}

As advances in procedural techniques extend intervention to increasingly complex populations, untangling the relationship between AS and HFpEF has never been more relevant. AS and HFpEF are likely to coexist frequently, interacting via common mechanistic pathways. The presence of HFpEF in AS impacts clinical presentation and outcomes,^{4,5} while even mild AS worsens prognosis in HFpEF.⁶ In addition, emerging data indicate that newer HFpEF-directed therapies may benefit selected patients with AS.^{7,8} In this review, we synthesize epidemiological, pathophysiological and clinical evidence on the AS-HFpEF overlap, and explore

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Nonstandard Abbreviations and Acronyms

AF	atrial fibrillation
AS	aortic stenosis
AVR	aortic valve replacement
HF	heart failure
HFpEF	heart failure with preserved ejection fraction
LV	left ventricular
LVEF	left ventricular ejection fraction
PCWP	pulmonary capillary wedge pressure
PH	pulmonary hypertension
PK	protein kinase
PLF-LG	paradoxical low-flow low-gradient
SAVR	surgical aortic valve replacement
TAVR	transcatheter aortic valve replacement

its prevalence, clinical implications, and therapeutic opportunities.

EPIDEMIOLOGY OF AS AND HFpEF

Shared Risk Factors

Calcific AS, characterized by fibrocalcific degeneration of a congenital bicuspid or tricuspid aortic valve, has now surpassed rheumatic heart disease as the predominant cause of AS in the developed world.² Accompanying this shift has been a change in the typical patient with AS. Once a condition of younger adults with prior rheumatic fever or congenital valve disease, AS is now predominantly a disease of the elderly, affecting an estimated 12% of individuals aged over 75 years.⁹ Beyond age, risk factors include hypertension, diabetes, obesity, smoking, hypercholesterolemia, elevated lipoprotein(a), and chronic kidney disease (Figure 1),^{2,10} of which diabetes,^{11,12} the metabolic syndrome,¹³ and hypertension¹⁴ additionally confer a greater risk of concomitant maladaptive cardiac remodeling.

HFpEF is likewise a disease of an aging population.¹ Defined by the presence of HF symptoms or signs with a left ventricular ejection fraction (LVEF) >50%, alongside structural and functional cardiac abnormalities, HFpEF affects 1% to 1.5% of adults and over 10% of those aged over 80 years.¹⁵ Risk factors include older age, hypertension, obesity, diabetes, the metabolic syndrome, chronic kidney disease, coronary artery disease, and atrial fibrillation (AF)—many of which overlap with or are common in, populations with AS (Figure 1).^{1,16} With continued population aging and the rising prevalence of associated comorbidities, the burden of both conditions is projected to increase.^{1,2}

Sex Differences

HFpEF has traditionally been considered a disease of women, who account for two-thirds of cases. Compared

with men, women exhibit smaller left ventricular (LV) cavities, more concentric remodeling, increased arterial elastance and wave reflection with aging, and a higher prevalence of coronary microvascular dysfunction—all factors predisposing to HFpEF.¹⁷ AS is more prevalent in men, driven by a 3:1 male predominance of congenital bicuspid aortic valve disease.¹⁸ However, this difference narrows with age, and women comprise the majority of AS patients over 75.⁹ Women with AS exhibit a distinct remodeling response to pressure overload that more closely resembles HFpEF, characterized by preserved LVEF, concentric remodeling, diastolic dysfunction, and elevated filling pressures.¹⁸ Thus, the AS-HFpEF overlap may be particularly relevant in older women.

Prevalence of AS-HFpEF Overlap

AS and HFpEF have traditionally been considered distinct entities, with guidelines emphasizing the need to exclude HFpEF masqueraders such as valvular disease before diagnosing HFpEF.¹ While appropriate in an era when AS predominantly affected younger patients, this framework is less applicable today, where the mean age of patients undergoing AVR is around 80 years and multimorbidity is the norm rather than the exception.² Given substantial overlap in risk factors, AS and HFpEF may frequently evolve in parallel rather than in isolation, particularly in older adults (Figure 1).

Although longitudinal studies are lacking, epidemiological data support this. Up to 30% of patients hospitalized with HFpEF have some degree of AS (Table 1), far exceeding rates in the general population.^{6,19–21} AS appears more strongly correlated with HFpEF than HF with reduced ejection fraction, perhaps reflecting greater overlap in risk factors and mechanisms.⁶ Similarly, HFpEF diagnostic scores identify HFpEF features in up to 20% of patients with moderate AS, increasing to almost half in severe AS (Table 2).^{4,5,22–26} As both conditions grow more prevalent, this overlap will be encountered with increasing frequency.

PATHOPHYSIOLOGY OF AS AND HFpEF

Beyond shared risk factors, AS and HFpEF also interact through several shared mechanisms. Both converge on a similar cardiac phenotype characterized by concentric remodeling, diastolic dysfunction, and elevated filling pressures, and both are influenced by comorbidities and systemic factors acting through pathways such as inflammation, fibrosis, coronary microvascular dysfunction, and altered cardiac metabolism and energetics (Figure 2; Table S1).

LV Structural and Functional Abnormalities in AS and HFpEF

While the primary abnormality in AS is valvular obstruction, the accompanying myocardial response is equally

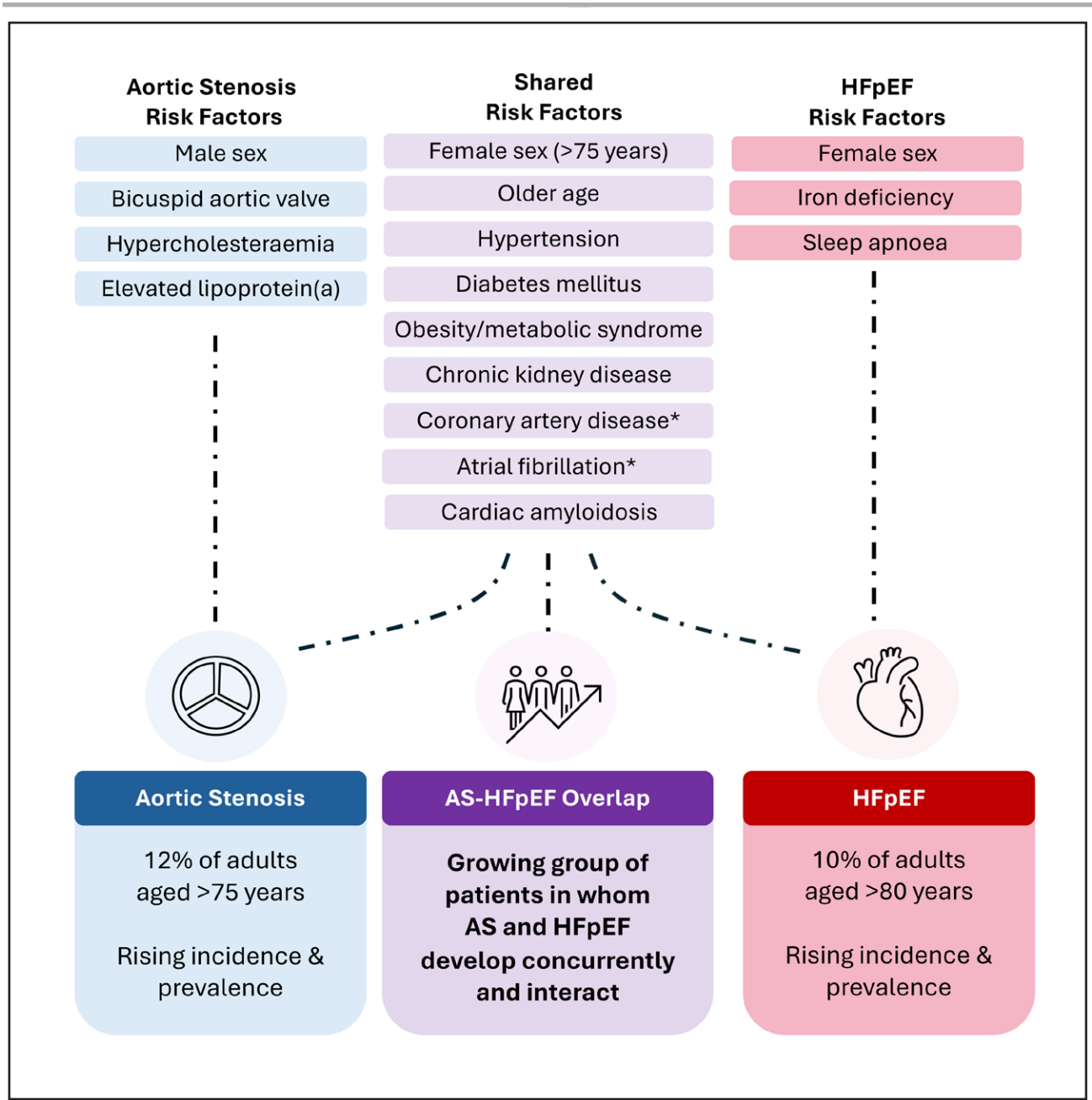


Figure 1. Shared risk factors for aortic stenosis (AS) and heart failure with preserved ejection fraction (HFpEF). AS and HFpEF are 2 of the most common cardiovascular conditions in older adults and share multiple risk factors that contribute to disease development and progression. As both conditions become more prevalent, a growing subset of patients present with overlapping features of both AS and HFpEF, complicating diagnosis and management. Asterisked conditions (*) denote HFpEF risk factors that are prevalent in populations with AS.

important. Chronic pressure overload initiates a complex process of hypertrophic remodeling (Figure 3A), producing a thick-walled and concentrically remodeled LV.²⁷ This initial compensatory response serves to preserve systolic wall stress and ejection fraction in the face of elevated LV afterload, but occurs at the cost of increasing LV stiffness, impaired relaxation, and ultimately diastolic dysfunction.^{27,28}

With disease progression, maladaptive changes such as cardiomyocyte apoptosis/necrosis,²⁹ coronary microvascular dysfunction,³⁰ myocardial fibrosis,²⁷ titin hypophosphorylation,³¹ and altered cardiac metabolism and energetics³² further compromise LV diastolic function. Subclinical systolic impairment is common, manifesting as reduced LV global longitudinal strain despite preserved LVEF.³³ As diastolic dysfunction worsens, LV filling becomes increasingly reliant on elevated filling pressures.³⁴ Importantly, it is this rise in filling pressures, rather than valve stenosis alone, that appears to be the dominant driver of dyspnea and HF symptoms in AS.³⁵

Table 1. Prevalence and Prognostic Impact of Aortic Stenosis in HFpEF Cohorts

References	Population and study design	N	Age, y	Female, %	Prevalence of AS in HFpEF	Prognostic impact of AS in HFpEF
Shah 2019 ¹⁹	Retrospective study of PARAGON-HF trial data	1097 (864 HFpEF)	74±8	53	Across all LVEF categories 10% had mild AS 2% had mod/severe AS	Any degree of AS associated with ↑HF hospitalization or CV mortality across all LVEF categories
Verbrugge 2021 ²⁰	Retrospective cohort study of consecutive patients hospitalized with HFpEF at a single center	370 (All HFpEF)	77±13	58	In HFpEF 30% had mild AS Mod/severe AS not assessed	Mild AS associated with ↑LV filling pressures, left shift of LV end-diastolic pressure-volume relationship and ↑mortality in HFpEF
Shahim 2023 ²¹	Prospective multicenter, observational study of patients enrolled in the ESC-HF-LT Registry (acute or chronic HF spanning 133 cardiology centers, from 21 countries)	15 216 (3120 HFpEF)	67 (58–76)	31	In HFpEF cohort Mild AS not assessed 6% had mod/severe AS In HFrEF cohort Mild AS not assessed 4% had mod/severe AS	Moderate/severe AS associated with ↑HF hospitalization or CV mortality across all LVEF categories
Sivaraj 2023 ⁶	Retrospective cohort study of patients ≥55 y in 4 US communities who were hospitalized with acute decompensated HF	3579 (1849 HFpEF)	74±11	56	In HFpEF cohort 14% had mild-mod AS 4% had severe AS In HFrEF cohort 8% had mild-mod AS 4% had severe AS	Any degree of AS associated with ↑mortality across all LVEF categories

AS indicates aortic stenosis; CV, cardiovascular; HF, heart failure; HFpEF, heart failure with preserved ejection fraction; HFrEF, heart failure with reduced ejection fraction; and LVEF, left ventricular ejection fraction.

HFpEF demonstrates many similarities to the pressure-overloaded AS heart. Concentric remodeling and increased LV wall thickness are typical.³⁶ Cardiomyocyte hypertrophy and fibrosis are observed in 90% of HFpEF biopsies.³⁷ Diastolic dysfunction resulting from both impaired LV relaxation and passive chamber stiffening is a central feature.³⁸ Many patients with HFpEF also display subclinical systolic dysfunction.³⁹ The culmination of these abnormalities is an increase in LA pressure, widely considered the hemodynamic hallmark of HFpEF, which drives pulmonary congestion, ventilatory inefficiency, and cardinal symptoms of dyspnea and exercise intolerance.⁴⁰

Exercise Hemodynamics in AS and HFpEF

Invasive exercise hemodynamic assessment is the gold standard for HFpEF diagnosis.¹ Many patients with HFpEF maintain normal filling pressures at rest but exhibit marked increases in pulmonary capillary wedge pressure (PCWP) and pulmonary artery pressures during exertion.⁴¹ Resting cardiac output is typically normal or slightly reduced but fails to augment with exercise, reflecting both reduced stroke volume reserve and chronotropic incompetence.⁴¹ HFpEF is confirmed by elevated resting (≥15 mmHg) or exercise (≥25 mmHg) PCWP, or a PCWP-cardiac output slope >2 mmHg/L per minute.¹

Exercise hemodynamics in AS are less well characterized but show notable parallels. Resting cardiac output is normal or slightly reduced, but the increase in stroke volume and cardiac output during exercise is blunted, reflecting the dual burden of a fixed valvular obstruction and reduced myocardial reserve due to cardiac remodeling.⁴² As in HFpEF, PCWP is normal or

modestly elevated at rest but increases steeply relative to cardiac output with exertion (Table 3).^{34,43–48,50} Interestingly, these hemodynamic abnormalities correlate more closely with age, comorbidities, and markers of LV remodeling than with conventional measures of AS severity such as valve area and gradient, highlighting the importance of the myocardial response in shaping the clinical phenotype.²²

A subset of patients with HFpEF and AS exhibit reduced resting stroke volume despite preserved LVEF.⁵¹ In AS, this low-flow state may mask the true severity of valvular obstruction, leading to low transvalvular gradients despite severe stenosis—so-called paradoxical low-flow low-gradient (PLF-LG) AS.⁵¹

Paradoxical Low-Flow Low-Gradient AS

Severe AS is defined as a peak aortic jet velocity ≥4 m/s, mean transvalvular gradient ≥40 mmHg, and aortic valve area <1.0 cm².⁵² However, some patients present with low-gradient AS—a small valve area consistent with severe AS, but a gradient <40 mmHg suggestive of non-severe disease. The most frequent cause of this is a low outflow state (stroke volume index <35 mL/m²), which may occur either with reduced (classical LF-LG AS) or preserved LVEF (PLF-LG AS).

PLF-LG AS affects 25% to 35% of patients with severe AS and preserved LVEF.⁵¹ Compared with high-gradient AS, these patients are older, more often female, and have a higher prevalence of typical HFpEF comorbidities including hypertension and AF (Table S2).⁵¹ They exhibit more concentric remodeling, fibrosis, and diastolic and systolic longitudinal dysfunction compared with other AS subtypes.^{53–55} They also frequently display increased

Table 2. Studies Applying HFpEF Risk Scores in Patients With Aortic Stenosis

References	Study type	Population	HFpEF score and distribution	High HFpEF score vs low-intermediate HFpEF score		
				Aortic valve parameters	Clinical features	Outcomes
Venema 2025 ⁵	Single center retrospective cohort study (2 separate cohorts evaluated)	Moderate AS and LVEF $\geq 50\%$ (n=469), median age ≈ 68 y, 39% female	H ₂ FPEF 0–5: 81% H ₂ FPEF 6–9: 9%	$\leftrightarrow$ AV area $\leftrightarrow$ AV MPG	$\uparrow$ LV mass index $\uparrow$ Left atrial volumes $\uparrow$ Symptom onset at lower aortic valve gradients	$\uparrow$ Progression to AVR over 5 y
		Severe AS undergoing TAVR and LVEF $\geq 50\%$ (n=601), median age ≈ 81 y, 60% female	H ₂ FPEF 0–5: 80% H ₂ FPEF 6–9: 20%	$\leftrightarrow$ AV area $\downarrow$ AV MPG	$\uparrow$ Filling pressures $\uparrow$ Left atrial volumes $\uparrow$ Pulmonary pressures $\uparrow$ MR and TR	$\uparrow$ NYHA III/IV symptoms post-TAVR $\uparrow$ NT-proBNP and left atrial dysfunction post-TAVR $\uparrow$ CV death post-TAVR
Hoedemakers 2024 ²²	Single center, prospective, observational	Moderate symptomatic AS, or severe asymptomatic AS and LVEF $\geq 50\%$ (n=206), mean age 73, 40% female	H ₂ FPEF 0–5: 80% H ₂ FPEF 6–9: 20%	$\leftrightarrow$ AV area $\leftrightarrow$ AV MPG	$\uparrow$ Diastolic dysfunction $\downarrow$ Global longitudinal strain $\uparrow$ Left atrial volumes/strain $\uparrow$ Pulmonary pressures $\uparrow$ MPAP/CO slope $\downarrow$ Right ventricular function $\downarrow$ Exercise capacity	Not assessed
Ishizu 2024 ²³	Single center, retrospective, observational	Severe AS undergoing TAVR and LVEF $\geq 50\%$ (n=1674), mean age 85, 72% female	H ₂ FPEF 0–5: 84% H ₂ FPEF 6–9: 16%	$\leftrightarrow$ AV area $\downarrow$ AV MPG $\uparrow$ PLF-LG AS	$\uparrow$ NYHA III/IV symptoms $\uparrow$ LV filling pressures $\uparrow$ Pulmonary pressures $\uparrow$ MR and TR	$\uparrow$ Mortality post-TAVR $\uparrow$ CV mortality or HF hospitalization post-TAVR
Akao 2023 ²⁴	Single center, retrospective, observational	Severe AS undergoing TAVR (n=244), mean age 86, 71% female	H ₂ FPEF 0–5: 89% H ₂ FPEF 6–9: 11%	$\leftrightarrow$ AV area $\downarrow$ AV MPG	$\uparrow$ Left atrial size $\uparrow$ Pulmonary pressures $\uparrow$ MR and TR	$\uparrow$ Mortality or HF hospitalization post-TAVR
Ludwig 2021 ²⁵	Two-center, retrospective, observational	Severe AS undergoing TAVR and LVEF $\geq 50\%$ (n=832), mean age 82, 57% female	H ₂ FPEF 0–5: 69% H ₂ FPEF 6–9: 31%	$\downarrow$ AV MPG $\uparrow$ PLF-LG AS	$\uparrow$ NYHA III/IV symptoms $\uparrow$ LV filling pressures $\uparrow$ Pulmonary pressures $\uparrow$ MR and TR	$\uparrow$ Mortality post-TAVR $\uparrow$ CV mortality or HF hospitalization post-TAVR
Seoudy 2022 ²⁶	Single center, retrospective, observational	Severe AS undergoing TAVR and LVEF $\geq 50\%$ (n=570), mean age 82, 60% female	HFA-PEFF <5: 58% HFA-PEFF ≥ 5 : 42%	$\leftrightarrow$ AV area $\leftrightarrow$ AV MPG	$\uparrow$ NYHA III/IV symptoms $\uparrow$ LV mass $\uparrow$ LV filling pressures $\uparrow$ Left atrial size $\uparrow$ Pulmonary pressures $\uparrow$ MR and TR $\uparrow$ Right ventricular dysfunction $\uparrow$ NT-proBNP	$\uparrow$ Post-TAVR PPM insertion $\uparrow$ Mortality or HF hospitalization post-TAVR
Kammerlander 2021 ⁴	Single center, retrospective, observational	Late post-SAVR (median 30 mo) and LVEF $\geq 50\%$ (n=415), mean age 72, 50% female	HFA-PEFF <5: 49% HFA-PEFF ≥ 5 : 51%	Not reported	Not reported	$\uparrow$ Mortality late post-SAVR

AS indicates aortic stenosis; AV, aortic valve; CV, cardiovascular; HF, heart failure; HFpEF, heart failure with preserved ejection fraction; LV, left ventricular; LVEF, left ventricular ejection fraction; MPG, mean pressure gradient; MPAP, mean pulmonary artery pressure; MR, mitral regurgitation; NYHA, New York Heart Association; PLF-LG AS, paradoxical low-flow low-gradient aortic stenosis; SAVR, surgical aortic valve replacement; TAVR, transcatheter aortic valve replacement; and TR, tricuspid regurgitation.

valvuloarterial impedance due to hypertension or increased systemic arterial afterload,^{53,55} a greater prevalence of mitral or tricuspid disease, and right ventricular dysfunction, which may further reduce LV outflow.⁵¹

Longitudinal studies indicate that many patients with PLF-LG AS develop concentric remodeling and impaired LV filling before progressing to severe AS, implying a cardiomyopathic process that coexists with, or predates, valvular disease.⁵⁴ PLF-LG AS has therefore been proposed to represent the intersection of 2 common diseases of the elderly: HFpEF and AS.⁵¹ Consistent with this, AVR alone appears insufficient to address the full spectrum of myocardial, vascular, and systemic abnormalities in PLF-LG

AS. While these patients still benefit from AVR, outcomes are inferior to those with classical high-gradient AS.⁵¹

Multichamber Cardiac Involvement

In both AS and HFpEF, elevated LV filling pressures are transmitted to the LA, resulting in LA remodeling, stiffness, and impaired reservoir function.^{34,56,57} LA dysfunction predisposes to AF, which is common in both conditions and contributes to worsening hemodynamics and LA mechanics in a vicious cycle.^{16,58} Functional atrial mitral regurgitation can develop as a consequence of annular dilatation and chronic AF.⁵⁹ LA abnormalities

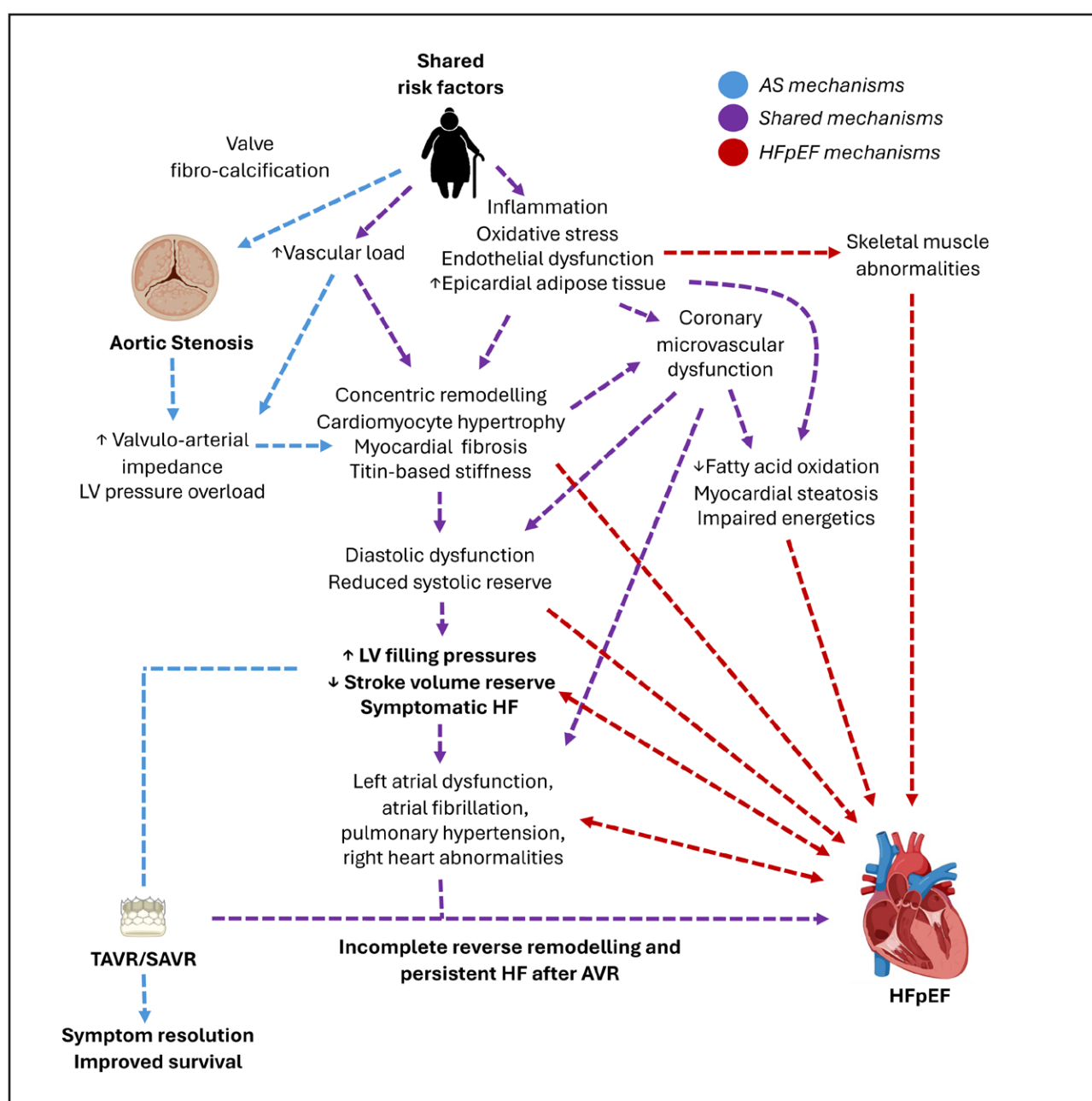


Figure 2. Shared mechanisms linking aortic stenosis (AS) and heart failure with preserved ejection fraction (HFpEF).

AS and HFpEF share risk factors and mechanisms that converge on a common cardiac phenotype characterized by concentric left ventricular (LV) remodeling, myocardial fibrosis, and titin-based myocardial stiffening, resulting in diastolic dysfunction, longitudinal systolic impairment, elevated LV filling pressures, and heart failure (HF). In both conditions, cardiac abnormalities arise not only from the hemodynamic load related to valvular and vascular factors but are also shaped by comorbidities and systemic processes such as inflammation, microvascular dysfunction, and cardiac metabolic and energetic impairment. AS and HFpEF may interact with and amplify one another, worsening the clinical phenotype in individuals with overlapping disease. Moreover, AS itself may induce HFpEF-like changes in cardiac structure and function that are essentially indistinguishable from primary HFpEF. When such cardiac remodeling is advanced, it may contribute to incomplete reverse remodeling and persistent HF symptoms following aortic valve replacement (AVR).

carry independent prognostic significance in both AS and HFpEF, and in a subset of HFpEF patients may occur even without overt LV abnormalities.^{56,60}

Pulmonary hypertension (PH) affects up to 75% of patients with severe AS and up to 83% of those with HFpEF.^{61,62} In most cases, PH is postcapillary, driven by backward transmission of elevated LA pressures.

However, an important minority develop combined pre- and postcapillary PH with elevated pulmonary vascular resistance and pulmonary vascular remodeling, associated with particularly poor outcomes.^{61,62}

Right ventricular dysfunction is also a poor prognostic marker.^{63,64} While most often developing in the context of chronic PH, it can occur without overt elevation in

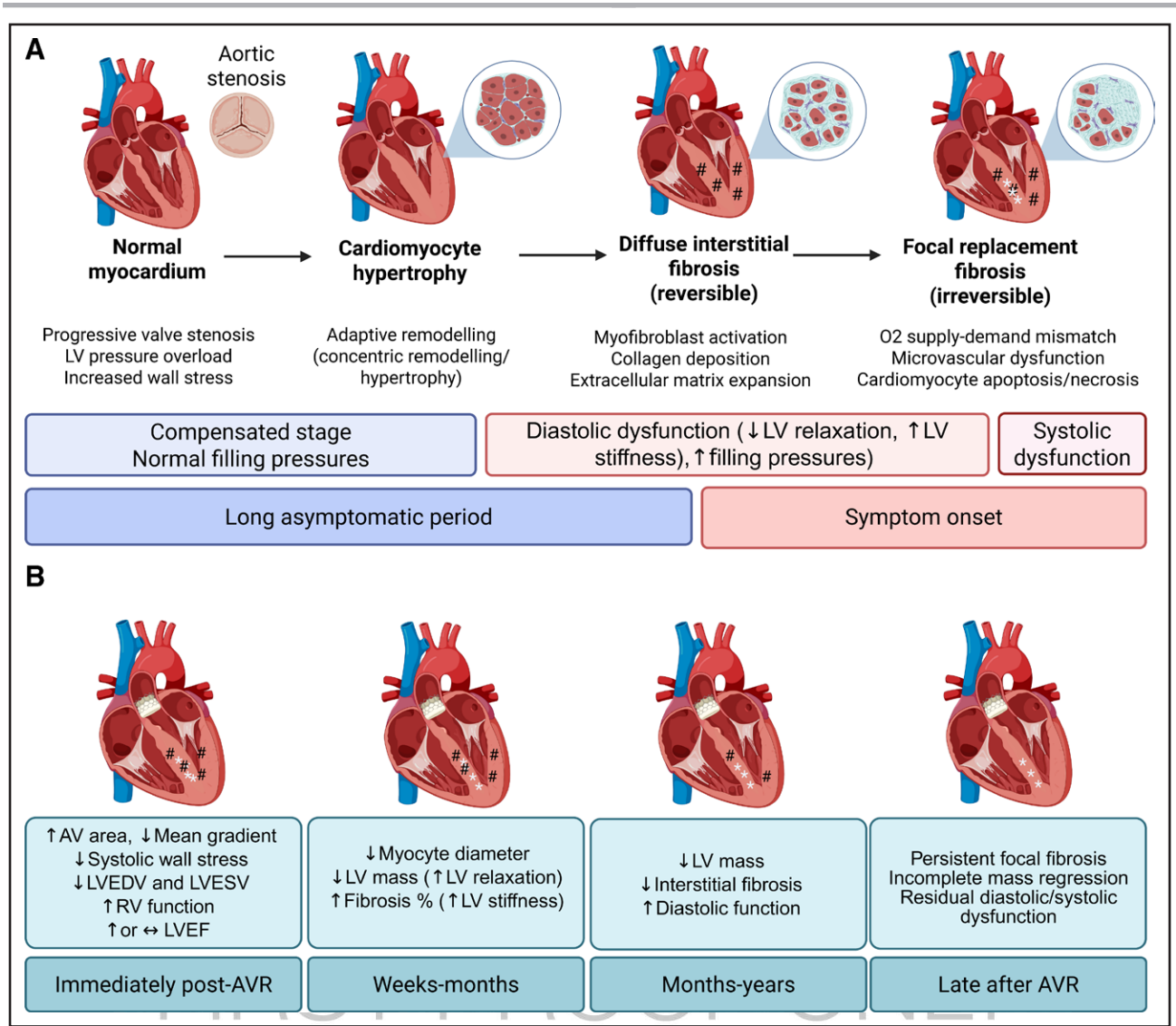


Figure 3. Simplified schematic of myocardial remodeling in aortic stenosis and reverse remodeling following aortic valve replacement (AVR).

A, Chronic pressure overload from aortic stenosis drives progression from adaptive concentric hypertrophy to diffuse interstitial fibrosis and, ultimately, irreversible replacement fibrosis. These changes are accompanied by worsening diastolic dysfunction, elevated left ventricular (LV) filling pressures, and eventual symptom onset. The trajectory is heterogeneous and influenced by both the degree of valve stenosis and systemic factors. **B**, AVR rapidly improves valvular hemodynamics (increased aortic valve area [AVA], decreased mean pressure gradient [MPG]) and reduces left ventricular (LV) afterload and systolic wall stress, with an early reduction in chamber volumes and improvement in right ventricular (RV) function. Regression of myocardial abnormalities is slower and often incomplete. Cardiomyocyte hypertrophy regresses early, with a corresponding reduction in LV mass and improved LV relaxation. However, diffuse fibrosis may increase relative to declining LV mass in the early post-AVR period, resulting in paradoxically increased chamber stiffness. Late after AVR, diffuse fibrosis regresses, but focal fibrosis largely persists. Residual abnormalities may contribute to persistent diastolic dysfunction and heart failure symptoms after AVR. Created in BioRender. Zhou, J. (2026) <https://BioRender.com/2dvqazr>.

pulmonary pressures.⁶⁴ This points to a more complex pathophysiology than backward transmission of left-sided pressures, with proposed mechanisms including systolic ventricular interaction or the influence of systemic factors that affect both ventricles.⁶⁵

Inflammation

Inflammation is believed to play a central role in HFpEF and is increasingly implicated in AS.⁶⁶ A widely accepted

paradigm of HFpEF proposes that comorbidities such as diabetes, obesity, and chronic kidney disease foster a systemic proinflammatory milieu that promotes widespread endothelial inflammation and coronary microvascular dysfunction, with hemodynamic load from arterial hypertension amplifying this cascade.⁶⁶

In AS, inflammation is a recognized driver of valve fibrocalcification,² but may also contribute to adverse LV remodeling. Pressure overload from aortic banding in animal models triggers proinflammatory signaling,

Table 3. Exercise Hemodynamics Before and After Aortic Valve Replacement in Severe Aortic Stenosis

References	Population	Before AVR				After AVR			
		Resting PCWP (mm Hg)	Exercise PCWP (mm Hg)	R-PCWP ≥15 mm Hg (%)	E-PCWP ≥25 mm Hg (%)	Resting PCWP (mm Hg)	Exercise PCWP (mm Hg)	R-PCWP ≥15 mm Hg (%)	E-PCWP ≥25 mm Hg (%)
Exercise hemodynamic assessment before AVR									
Dalsgaard 2009 ⁴³	Severe AS (n=28), mean age 70 y, 61% male	18±8	39±10	61%	92%				
Dalsgaard 2014 ⁴⁴	Severe symptomatic AS (n=44), mean age 70 y, 77% male	17±7	39±10	...	...				
Christensen 2016 ³⁴	Severe asymptomatic AS with LVEF ≥50% (n=39), mean age 73 y, 77% male	12±3	31±5	13%	85%				
Dobarro 2020 ⁴⁵	Moderate or severe asymptomatic AS with LVEF ≥50% (n=33), mean age 74 y, 91% male	14±5	22±8	...	...				
Kvaslerud 2022 ⁴⁶	Severe asymptomatic AS with LVEF ≥50 (n=50), mean age 70 y, 64% male	11±4	27±7	20%	66%				
Al-Abcha 2025 ⁴²	Moderate+AS with LVEF ≥50% (n=34), mean age 77 y, 50% male	14 (11–21)	29±9	41%	70%				
Exercise hemodynamic assessment after AVR									
Monrad 1988 ⁴⁷	Severe AS, 10 y post-SAVR (n=12), mean age 52 y, 75% male					9±3	24±10	25%	42%
Carter-Stoch 2018 ⁴⁸	Severe AS, 1 y post-SAVR (n=37), mean age 72 y, 73% male					10±3	26±7	11%	32%
Exercise hemodynamic assessment before and after AVR									
Inselmann 2000 ⁴⁹	Severe AS, before and 2.8 y post-SAVR (n=12), mean age 54 y, 71% male	17±9	...	33%	...	12±5	32±9	43%	71%

AS indicates aortic stenosis; AVR, aortic valve replacement; E-PCWP, pulmonary capillary wedge pressure during exercise; LVEF, left ventricular ejection fraction; PCWP, pulmonary capillary wedge pressure; R-PCWP, pulmonary capillary wedge pressure at rest; and SAVR, surgical aortic valve replacement.

myocardial inflammation, and fibrosis.⁶⁷ In humans, inflammatory plasma proteins are differentially expressed in patients with AS with reduced versus preserved LV global longitudinal strain.⁶⁸ Diabetes, in particular, appears to exacerbate this proinflammatory milieu and is associated with more pronounced fibrosis and diastolic dysfunction in AS.^{11,12}

Myocardial Fibrosis

Myocardial fibrosis is another shared feature. In early AS, pressure overload activates myofibroblasts, promoting diffuse but reversible collagen deposition (reactive interstitial fibrosis).²⁷ With disease progression, areas of cardiomyocyte injury are replaced by irreversible scar tissue (focal replacement fibrosis).²⁷ Both forms of fibrosis alter myocardial viscoelastic properties, increasing LV stiffness and contributing to diastolic dysfunction.⁶⁹ In AS, fibrotic burden correlates with higher filling pressures, greater symptom severity, delayed reverse remodeling post-AVR, and worse outcomes.^{69–71}

More than 90% of patients with HFpEF have evidence of interstitial fibrosis on endomyocardial biopsy, although such accumulation is also common in the aging heart.³⁷ Cardiac magnetic resonance studies similarly demonstrate increased focal and diffuse fibrosis in HFpEF compared with controls.⁷² The underlying mechanisms are incompletely understood but are thought to involve inflammation triggered by both metabolic and hemodynamic stress.⁶⁶

Titin Abnormalities

The giant myofilament protein titin is a key determinant of intrinsic cardiomyocyte stiffness, with its mechanical properties modulated by isoform expression and phosphorylation by protein kinases (PK)A and PKG.⁶⁶ In HFpEF, altered nitric oxide-cyclic guanosine monophosphate-PKG signaling reduces titin phosphorylation and increases cardiomyocyte resting tension, contributing to elevated LV stiffness.^{66,73} Modification of titin phosphorylation is 1 mechanism through which SGLT2 inhibitors may exert favorable effects in HFpEF.⁷⁴

Emerging data suggest titin abnormalities also contribute to diastolic dysfunction in AS. Endomyocardial biopsies demonstrate decreased titin phosphorylation at PKA and PKG sites,^{12,31,73} most evident in diabetic patients, in whom abnormal nitric oxide-cyclic guanosine monophosphate-PKG signaling, reduced titin phosphorylation, and increased cardiomyocyte resting tension are observed relative to nondiabetic patients with AS.^{11,12}

Coronary Microvascular Dysfunction

Coronary microvascular dysfunction affects up to 50% of patients with severe AS³⁰ and up to 75% of those with HFpEF.⁷⁵ In AS, contributors include microvascular rarefaction, perivascular fibrosis, endothelial dysfunction, reduced vasodilatory reserve, shortened diastolic time fraction, decreased coronary perfusion pressure, and extravascular compression from elevated LV end-diastolic pressures and wall stress.³⁰ In HFpEF, inflammation-mediated endothelial dysfunction and microvascular rarefaction are thought to be the key drivers.^{75,76} Coronary microvascular dysfunction, in turn, impairs myocardial oxygen delivery, blunts diastolic and contractile reserve, and limits exercise tolerance.⁷⁷ Recurrent microvascular ischemia may also drive cardiomyocyte injury and fibrosis, contributing to adverse remodeling.⁷⁶

Cardiac Metabolism and Energetics

The normal adult heart primarily relies on mitochondrial fatty acid oxidation for energy. HFpEF hearts exhibit reduced fatty acid uptake and oxidation, along with an inability to upregulate alternative pathways, resulting in a metabolically inflexible heart vulnerable to energetic stress.⁷⁸ This is accompanied by myocardial triglyceride accumulation (steatosis) and a reduced phosphocreatine/ATP ratio detectable on cardiac magnetic resonance spectroscopy.⁷⁹ Similar abnormalities are observed in AS,³² particularly in patients with concomitant diabetes or the metabolic syndrome, although AS hearts seem to compensate partially through increased glucose and ketone utilization.⁸⁰

Epicardial Adipose Tissue

Epicardial adipose tissue is a metabolically active tissue with potential relevance to both HFpEF and AS.⁸¹ Epicardial adipose tissue volume is higher in HFpEF than in controls and may contribute to adverse hemodynamics and outcomes, though this remains controversial. Proposed mechanisms include secretion of proinflammatory and profibrotic cytokines, altered metabolic signaling, autonomic dysregulation, and direct mechanical effects on diastolic filling.⁸¹ Epicardial adipose tissue may similarly influence disease progression

in AS.^{82,83} Patients with AS exhibit increased epicardial adipose tissue thickness compared with controls, correlating with elevated proinflammatory cytokine levels, increased valve stenosis and calcification, and maladaptive remodeling.

Cardiac Amyloidosis

Distinct from the mechanisms above, cardiac amyloidosis has emerged as an important contributor to both AS and HFpEF.⁸⁴ Amyloid fibril deposition may involve both the myocardium (causing wall thickening and diastolic dysfunction) and the aortic valve (contributing to leaflet calcification and stenosis).⁸⁴ Advances in noninvasive imaging have improved recognition, and a substantial proportion of patients previously labeled as isolated AS or HFpEF are now recognized to have amyloid. Transthyretin amyloidosis is the most common subtype in older adults, affecting 4% to 29% of patients with AS⁸⁵ and 13% to 17% of those with HFpEF.⁸⁴ These patients more frequently present with PLF-LG AS and experience worse outcomes than either condition alone.^{86,87} Amyloidosis should therefore be considered in the differential diagnosis of the AS-HFpEF overlap.

AS and HFpEF Overlap

The conceptual framework of AS has evolved from a disorder defined solely by valvular obstruction to one in which the myocardial response is equally important. Nonvalvular abnormalities such as fibrosis, diastolic dysfunction, and elevated filling pressures correlate more closely with symptom severity and prognosis than indices of valve severity alone, and their persistence after AVR is a strong determinant of outcomes. This recognition has fueled interest in staging systems that classify AS by the extent of extra-valvular involvement. One widely cited framework by Genereux and colleagues⁶⁴ describes progression from isolated valvular obstruction (Stage 0) to LV remodeling, LA dysfunction, PH, and right ventricular impairment (Stage 4), each with increasing mortality.

While intuitive, such staging systems do not fully capture the complexity of AS pathophysiology. Extra-valvular abnormalities do not always evolve in a predictable sequence beginning at the valve, display marked heterogeneity, and correlate only moderately with valve stenosis severity.⁶⁴ These observations reflect the combined influence of pressure overload, systemic factors, and comorbidities on shaping the AS phenotype. Importantly, features commonly attributed to advanced AS may not be consequences of AS alone but rather manifestations of coexisting conditions such as HFpEF that arise through shared mechanistic pathways.

A central challenge is distinguishing AS-related cardiomyopathy with preserved ejection fraction, where a HFpEF-like phenotype arises secondary to chronic

pressure overload, from coexisting primary HFpEF, which may precede AS or develop alongside it through shared risk factors. In practice, this distinction is rarely clear, and no validated approach exists to differentiate these entities at a patient level. In many patients, both processes likely coexist and interact, amplifying disease expression and symptom burden, and contributing to heterogeneity in post-AVR outcomes (Figure 2).

CLINICAL CONSEQUENCES OF AS-HFpEF OVERLAP

HF After AVR

HF is the most common cause of rehospitalization after AVR and is associated with increased long-term mortality.⁸⁸ Randomized trials report HF hospitalization rates of up to 13% within the first year after AVR,⁸⁹ with rates of up to 24% in registries.⁹⁰ Most affected patients have preserved LVEF and satisfactory valve parameters, such that post-AVR HF largely represents the persistence or development of a HFpEF phenotype.

Despite this, HFpEF after AVR remains underrecognized. In one of the few studies to evaluate its prevalence, Kammerlander and colleagues assessed 415 patients at a median of 30 months after surgical AVR (SAVR).⁴ Strikingly, 64% met criteria for HFpEF based on symptoms, echocardiographic data, and elevated natriuretic peptide levels.⁴ These patients had worse survival than those without HFpEF, with outcomes comparable to conventional HFpEF cohorts, yet only 12% had been diagnosed before study enrollment.

A small number of invasive hemodynamic studies support this observation (Table 3). Across 3 studies (n=56) evaluating exercise hemodynamics 1 to 10 years after SAVR, 11% to 43% of patients demonstrated elevated PCWP at rest, increasing to 32% to 71% during exercise, technically meeting gold-standard hemodynamic criteria for HFpEF.^{47–49} Notably, these hemodynamic abnormalities correlated poorly with prosthetic valve function and were more closely associated with preexisting diastolic dysfunction, coronary artery disease, LA dilatation, and myocardial fibrosis.^{48,49,91} Comparable invasive hemodynamic data are lacking in contemporary post-TAVR populations, in whom the burden of persistent HFpEF physiology may be even greater given their older age and higher comorbidity burden.

Persistent Extra-Valvular Abnormalities after AVR

The persistence of a HFpEF phenotype post-AVR is in part explained by heterogeneity in the myocardial response to intervention. Although AVR effectively relieves valvular obstruction, many patients exhibit persistent nonvalvular abnormalities even years after intervention.

Typical stages of reverse remodeling post-AVR are presented in Figure 3B. Relief of valvular stenosis produces an immediate reduction in LV pressure overload and systolic wall stress.⁹² This is accompanied by early regression of cardiomyocyte hypertrophy and LV mass, and corresponding improvement in LV diastolic function,^{27,93} with the greatest reduction in the first month.⁹⁴ Further mass regression occurs over months to years, although many patients exhibit increased LV mass relative to age- and sex-matched controls even late after AVR.⁴⁷

Myocardial fibrosis follows a more complex trajectory. Diffuse fibrosis regresses more slowly than cellular hypertrophy and, in the early post-AVR period, may paradoxically increase when indexed to a rapidly shrinking LV mass, transiently increasing chamber stiffness and diastolic dysfunction.^{27,93,95} Over time, diffuse fibrosis regresses, while focal fibrosis remains largely irreversible, providing a substrate for persistent diastolic dysfunction and HFpEF-like physiology.^{27,93}

Diastolic dysfunction improves in two-thirds of patients in the first month post-AVR, but persists or worsens in an important subset.²⁸ PH demonstrates similar heterogeneity: isolated postcapillary PH exhibits the greatest improvement, while precapillary or combined pre- and postcapillary PH is less reversible.⁶¹ Using the Genereux staging system, this variability is observed across the spectrum of extra-valvular changes. At 1 year post-AVR, only 15.6% had improvement in cardiac damage stage, while 26.5% had worsened.⁶⁴

The extent of reverse remodeling post-AVR is strongly prognostic. Patients with persistent hypertrophy, fibrosis, diastolic dysfunction, and non-LV abnormalities experience greater symptom burden, poorer functional capacity, more HF, and poorer survival.^{28,64,93} While often interpreted as evidence of advanced AS, these findings likely reflect a combination of factors: pressure overload-induced cardiac remodeling that has progressed beyond reversibility in some patients, and coexisting HFpEF or other cardiac conditions in others, accounting for wide variability in post-AVR trajectories.

Prognostic Impact of HFpEF in AS

With growing recognition of the AS-HFpEF overlap, several studies have applied HFpEF risk scores to AS cohorts to better define the prognostic impact of combined disease (Table 2).^{22–26} These tools offer a structured approach to estimating the likelihood of HFpEF among patients with dyspnea, although notably they were derived in populations without underlying valvular disease. The H₂FPEF score includes 6 variables (hypertension, obesity, AF, pulmonary artery systolic pressure, age, E/e') with scores ≥6 indicating a ≥90% likelihood of HFpEF.¹ The HFA-PEFF score integrates functional, morphological, and biomarker parameters, with scores >5 representing high HFpEF probability.¹

Among patients with moderate AS, 10% to 20% were found to have a high score-based HFpEF probability.^{5,22} These individuals developed HF symptoms earlier and at lower valve gradients, exhibited more pronounced functional limitation and exercise hemodynamic abnormalities, and more often progressed to needing AVR over a 5-year period.^{5,22} In severe AS, an even greater proportion were classified as having a high HFpEF probability,^{5,23–26} which in turn was predictive of more advanced extra-valvular cardiac damage, a higher prevalence of PLF-LG AS, greater symptom burden, and poorer post-AVR outcomes.⁵

Because these tools were developed in populations without valvular disease, their application in AS warrants careful interpretation. Many of the score components are themselves consequences of AS-related cardiac remodeling, and their ability to distinguish primary cardiometabolic HFpEF from pressure overload-induced HFpEF-like physiology is therefore limited. Nevertheless, higher scores consistently identify patients with greater symptom burden, more advanced remodeling, and poorer outcomes, suggesting that the presence of a HFpEF phenotype (irrespective of etiology) carries prognostic weight. Further work is needed to determine whether distinct phenotypes carry different prognostic or therapeutic implications.

Prognostic Impact of AS in HFpEF

Finally, the AS-HFpEF interplay appears to be bidirectional. Between 6% and 30% of HFpEF patients exhibit some degree of AS (Table 1), with any degree of AS associated with a higher risk of HF hospitalization and mortality.^{6,19–21} In addition, patients with HFpEF with even a mild degree of AS exhibit higher LV filling pressures and a leftward shift of the LV end-diastolic pressure-volume relationship compared with patients without AS,¹⁹ suggesting that even modest increases in afterload from a stenotic valve could unfavorably interact with an already stiff, noncompliant HFpEF ventricular.²⁰

IMPLICATIONS FOR CLINICAL PRACTICE

The data support a paradigm in which AS and HFpEF are not isolated disease processes but instead frequently coexist and interact in ways that amplify clinical burden and attenuate treatment response. In this context, it is perhaps helpful to consider AS and HFpEF as occupying a spectrum of overlap, with patients exhibiting varying degrees of functional limitation driven to differing extents by each condition (Figure 4). An individual's position along this spectrum is determined by a dynamic interplay of valvular and vascular load, myocardial structure and function, extracardiac influences, and comorbidities. This framework helps explain why symptoms

and hemodynamics vary so widely among patients with similar degrees of valve stenosis, and why post-AVR outcomes are so heterogeneous.

Diagnostic Considerations

Clinicians should maintain a high degree of suspicion for coexisting HFpEF in AS, particularly among older adults, women, and those with multiple comorbidities. Recognizing this overlap is critical, as these patients are at heightened risk of persistent symptoms and rehospitalization post-AVR. Risk scores such as H₂FPEF and HFA-PEFF may identify AS patients with a greater likelihood of HFpEF, though their performance in this setting remains incompletely validated. Development of an AS-specific risk stratification tool represents an important unmet need.

In patients with both AS and HFpEF, and especially PLF-LG AS, cardiac amyloidosis should be considered.^{85,87} In high-risk individuals, diphosphonate scintigraphy and serum/urine light-chain testing are warranted,⁵² particularly when red-flag features are present (eg, carpal tunnel syndrome, lumbar spinal stenosis, biceps tendon rupture, polyneuropathy, conduction abnormalities, disproportionate HF or natriuretic peptide elevation, and predominant right ventricular dysfunction).⁸⁴ Observational data suggest that patients with severe AS and cardiac amyloidosis still benefit from AVR, with the best outcomes observed in those treated with both AVR and ATTR-specific therapy.⁹⁶ Thus, disease-modifying therapy for amyloidosis should be initiated alongside AVR in these patients.

Confirmation of AS Severity

Clarifying the dominant pathology among patients with features of both AS and HFpEF is challenging. HF symptoms are subjective, multifactorial, and may arise from either condition to varying extents. No validated tools currently determine whether symptoms are primarily driven by valvular obstruction, HFpEF, or both. Until such evidence emerges, a careful and accurate assessment of AS severity is crucial, as this remains the principal determinant of whether patients proceed to AVR under current guidelines.⁵²

In patients with discordant echocardiographic measurements or when symptoms are disproportionate to AS severity, effort should be made to exclude measurement errors. Hypertension can result in underestimation of AS severity, and repeat assessment after blood pressure is controlled is recommended.¹⁴ When uncertainty persists, cardiac computed tomography calcium scoring of the aortic valve provides a useful adjunct.⁵² Sex-specific calcium score thresholds have been incorporated into guidelines and can help confirm or refute the presence of severe AS, particularly in LF-LG or discordant cases.⁵²

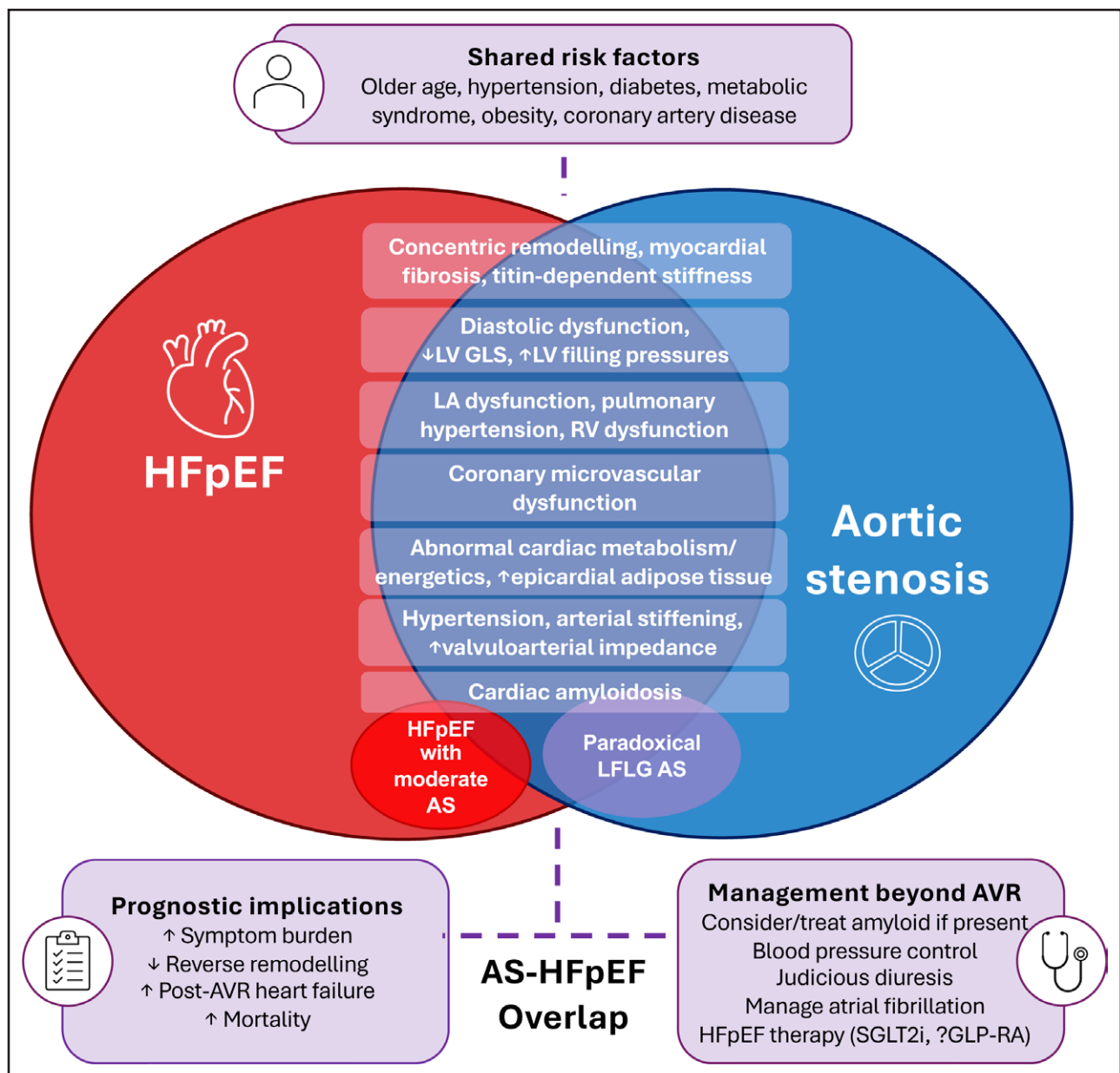


Figure 4. Aortic stenosis (AS) and heart failure with preserved ejection fraction (HFpEF) share overlapping risk factors, mechanisms, and frequently coexist, particularly in older patients with comorbidities.

Although traditionally considered distinct, these conditions often overlap. AS and HFpEF may be best conceptualized as existing along a spectrum, with individual patients exhibiting varying degrees of cardiac remodeling, hemodynamic derangement, and functional impairment driven to differing extents by each condition. While many patients present with isolated AS or HFpEF, others fall within an intermediate zone of overlap, characterized by a higher symptom burden, more frequent heart failure both before and after aortic valve replacement (AVR), and worse outcomes. A combined management approach that considers both valvular and myocardial contributions is essential.

Management of Comorbidities

Blood pressure management is important in both AS and HFpEF. In HFpEF, guidelines suggest targeting a systolic blood pressure of <130 mm Hg.¹ Drugs that target the renin-angiotensin-aldosterone system are preferred, while β -blockers are not recommended due to their negative chronotropic effects.⁹⁷

Although antihypertensives were historically avoided in AS due to concerns about precipitating hemodynamic

instability, contemporary data demonstrate that uncontrolled hypertension exacerbates LV hypertrophy, fibrosis, and symptom burden.¹⁴ Consequently, treatment of hypertension is advised with a more conservative target of <140/90 mm Hg.^{14,52} ACE inhibitors and angiotensin receptor blockers are preferred, with some evidence they may attenuate disease progression and cardiac remodeling in AS.¹⁴

AF is prevalent in both conditions and is associated with increased filling pressures and functional

impairment.^{16,58} AF can also contribute to a low-flow state and is more prevalent in PLF-LG AS.⁵¹ In HFpEF, emerging data point to a greater role for rhythm control, with a recent randomized study demonstrating that catheter ablation improved exercise capacity and quality of life among patients with AF and HFpEF.¹ Whether similar benefits extend to AS patients with AF and a HFpEF phenotype is unknown.

Suitability and Timing of AVR

Although AS patients with HFpEF experience poorer outcomes than those without HFpEF, data still support a clear symptomatic and survival benefit from AVR versus medical therapy.⁵² Thus, AVR should be offered to patients with confirmed severe symptomatic AS, including those with HFpEF, provided overall procedural risk and functional status are acceptable. In patients with significant extra-valvular cardiac involvement or a high probability of coexistent HFpEF, it is critical that preprocedural counseling address the possibility of residual symptoms post-AVR and the need for ongoing HF management.

Recognition that adverse cardiac remodeling can develop before symptom onset has challenged the paradigm of deferring AVR until the onset of severe symptomatic AS.⁹⁸ By the time symptoms manifest, patients may already have established fibrosis, diastolic dysfunction, atrial myopathy, and pulmonary vascular remodeling. Earlier intervention may offer an opportunity to attenuate progression of these abnormalities before they become only partially reversible.⁹⁸ Randomized trials in asymptomatic severe AS have demonstrated reductions in HF hospitalization with early AVR compared with watchful waiting, representing important emerging evidence in support of earlier intervention.⁹⁹ These data have informed ESC/EACTS guideline updates supporting AVR in selected asymptomatic patients.⁵² Ongoing trials (EASY-AS [REGISTRATION: URL: <https://www.clinicaltrials.gov>; Unique identifier: NCT04204915] and DAN-AVR [REGISTRATION: URL: <https://www.clinicaltrials.gov>; Unique identifier: NCT03972644]) are expected to further refine patient selection and timing.

Moderate symptomatic AS represents another evolving area.⁹⁸ This group likely encompasses a significant proportion of the AS-HFpEF overlap, where symptoms despite only moderate stenosis may be driven less by the valve lesion than by coexisting myocardial and systemic abnormalities. In such patients, even modest pressure overload may be poorly tolerated by a noncompliant HFpEF ventricle, such that AVR could confer disproportionate hemodynamic benefit.²⁰ In addition, the onset of symptoms at lower valve gradients may itself identify a subgroup with heightened susceptibility to pressure overload and adverse cardiac remodeling, in whom continued stenosis may accelerate deterioration. Equally, however, symptoms may be predominantly attributable to

HFpEF or other comorbidities, limiting the benefit of AVR alone.^{3,22}

Observational studies suggest potential benefit of AVR in moderate symptomatic AS.⁹⁸ However, the only randomized trial to date (TAVR UNLOAD) did not demonstrate superiority of transcatheter AVR (TAVR) over medical therapy in moderate AS and reduced LVEF, although quality-of-life improvements were observed.¹⁰⁰ Importantly, patients with preserved LVEF were excluded. Ongoing trials including PROGRESS (REGISTRATION: URL: <https://www.clinicaltrials.gov>; Unique identifier: NCT04889872) and EVOLUT EXPAND II (REGISTRATION: URL: <https://www.clinicaltrials.gov>; Unique identifier: NCT05149755) are expected to clarify the role of earlier intervention across a wider spectrum of patients with moderate AS, including those with preserved LVEF.

Procedural Considerations

In the setting of concentric remodeling and diastolic dysfunction, patients with AS and HFpEF may be particularly vulnerable to post-AVR complications such as patient-prosthesis mismatch, paravalvular leak, AF, and permanent pacemaker insertion.⁵¹ Whether TAVR or SAVR is preferable remains unclear. Observational studies suggest TAVR may be associated with better outcomes compared with SAVR in AS and concomitant HF,¹⁰¹ and particularly in cardiac amyloidosis.¹⁰² Prospective studies are required to evaluate the impact of prosthesis type and procedural approach on hemodynamics, reverse remodeling, and long-term outcomes in combined AS and HFpEF.

Paradoxical LF-LG AS

Evidence to guide management of PLF-LG AS is limited and largely derived from retrospective studies, which generally support a survival benefit with AVR compared with medical therapy.¹⁰³ A post hoc analysis of the PARTNER I randomized trial demonstrated lower mortality with TAVR versus conservative management in PLF-LG AS, although this was not a prespecified analysis.¹⁰⁴ A more recent prospective observational study reported improved outcomes with early AVR versus conservative management, with a signal for benefit of TAVR over SAVR.¹⁰⁵ Guidelines generally recommend AVR in symptomatic PLF-LG AS following careful confirmation of AS severity.⁵²

Post-AVR Management

Patients with combined AS and HFpEF remain at increased risk of HF hospitalization after AVR and should be closely monitored for evidence of congestion. Fluid overload in the early post-TAVR period is associated with worse outcomes, whereas proactive decongestive

treatment after TAVR appears to improve outcomes and quality of life.¹⁰⁶ Early follow-up post-AVR is recommended to assess for HF symptoms and to ensure medical therapy is optimized per current guidelines.

HFpEF-Directed Treatment

Diuretics remain a cornerstone of HFpEF therapy and should be considered in AS patients with fluid overload,¹ though applied judiciously in those with severe AS, concentric geometry, and a small LV cavity, who are likely to be preload-dependent.¹⁴ Mineralocorticoid receptor antagonists may reduce HF hospitalization in HFpEF and are often used.¹ However, the only randomized trial evaluating mineralocorticoid receptor antagonists in AS found no benefit in attenuating myocardial remodeling or valve stenosis progression in asymptomatic moderate-to-severe disease.¹⁰⁷

SGLT2 inhibitors reduce cardiovascular death and HF hospitalization in HFpEF and now carry a Class I recommendation.⁷⁴ However, all HFpEF trials to date have excluded patients with significant valvular disease. The DAPA-TAVI trial provides the first randomized evidence supporting SGLT2 inhibitor initiation in severe AS.⁷ In this study, dapagliflozin initiated shortly after TAVR reduced the composite of death or worsening HF by 28% in AS patients considered to be at high risk of HF events. Although not restricted to HFpEF, the inclusion criteria (prior HF decompensation, plus diabetes, chronic kidney disease, or LVEF $\leq 40\%$) enriched for a trial population in which HFpEF was prevalent (mean LVEF 55%; only 17% with LVEF $\leq 40\%$). The mechanisms underlying SGLT2 inhibitor benefit after AVR are incompletely understood. Early separation of event curves in DAPA-TAVI suggests a predominant diuretic/natriuretic effect, though actions on inflammation, fibrosis, metabolism, and titin phosphorylation have been proposed.⁷⁴ Understanding these mechanisms will be essential in determining who would benefit most from therapy.

The optimal timing of SGLT2 inhibitor initiation in AS also remains uncertain. While DAPA-TAVI evaluated postprocedural initiation, observational data suggest potential advantages with earlier use. In patients with nonsevere AS, SGLT2 inhibitor therapy has been associated with slower progression to severe AS,⁸ while a propensity-matched cohort of >21 000 patients with AS linked SGLT2 inhibitor use with lower mortality and reduced need for AVR, albeit without systematic echocardiographic follow-up.¹⁰⁸ Only 1 small study (n=40) has evaluated use in severe symptomatic AS before AVR, showing that empagliflozin initiated in the 6 months before AVR was associated with reduced HF hospitalization and death, without an increase in adverse events.¹⁰⁹ Pending robust prospective data, cautious pre-AVR initiation may be considered with

close monitoring for volume depletion and hypotension. Ongoing randomized trials will further define the role and timing of SGLT2 inhibitors across the AS spectrum (Table 4).

Finally, glucagon-like peptide-1 receptor agonists (GLP-1RAs) have emerged as promising agents for the obese/cardiometabolic HFpEF phenotype, with effects extending beyond weight loss to include improvements in diastolic function, LV mass, myocardial metabolism, neurohormonal activation, and inflammation.¹¹⁰ Data in AS are limited. In a recent propensity-matched study, GLP-1RA use following TAVR was associated with reduced mortality, providing a signal for possible benefit.¹¹¹ The REVERSE-TAVR trial (REGISTRATION: URL: <https://www.clinicaltrials.gov>; Unique identifier: NCT07090343), evaluating pre-TAVR semaglutide in patients with AS with obesity or cardiometabolic disease, will provide further insight.

FUTURE DIRECTIONS

Despite growing recognition of the AS-HFpEF overlap, several challenges remain. Chief among these is untangling, within an individual patient, the relative contribution of each condition to symptoms and outcomes. This distinction is central to decisions regarding the suitability and timing of AVR, the expected post-AVR trajectory, and the role of adjunctive therapy. To date, progress has been hampered by the paradigm that these conditions are separate and unrelated, with most AS studies excluding HFpEF and vice versa.

Longitudinal studies following patients from early disease through the development of symptomatic AS, HFpEF, or both would help determine whether HFpEF evolves as a consequence of progressive valvular obstruction, arises independently through shared risk factors, or reflects a bidirectional process in which 1 accelerates the other. Among those with coexisting disease, integrating structural, functional, biomarker, and invasive hemodynamic data before and after AVR could help delineate phenotypes within this continuum and identify predictors of persistent HF.

Medical therapy in AS requires further investigation. While previous efforts have focused on modifying valve fibrocalcification, a more accessible target may be the myocardial remodeling that accompanies AS. Given the mechanistic convergence between AS and HFpEF, therapies proven effective in HFpEF hold particular promise. The DAPA-TAVI trial provides early proof of concept, with multiple ongoing studies expected to further define their role (Table 4). A key question is when such therapies should be initiated. Pre-AVR priming has been proposed as a strategy to enhance reverse remodeling and recovery post-AVR, but prospective studies are needed to define efficacy, optimal timing, and safety.

Table 4. Randomized Trials of SGLT2 Inhibitors and Incretin-Based Therapies in Aortic Stenosis

Trial (NCT)	Study design	Intervention	Population	Primary endpoint	N	Status
DAPA-TAVI (NCT04696185)	Randomized, open-label blinded end point trial	Dapagliflozin 10 mg daily initiated post-TAVR	Severe AS undergoing TAVR	Composite of death or worsening of heart failure (hospitalization or urgent visit) at 1 y	1222	Complete Significant reduction in primary outcome (HR, 0.72 [95% CI, 0.55–0.95])
ENAVO-TAVR (NCT05672836)	Randomized, double-blinded, placebo-controlled trial	Enavogliflozin 0.3 mg daily initiated post-TAVR	Severe AS undergoing TAVR LVEF $\geq$ 40% One of: LVH or LAE or $\uparrow$ NT-proBNP	Composite of death, nonfatal myocardial infarction, nonfatal stroke and heart failure hospitalization at 1 y	1041	Recruitment ongoing
EMPAVR (NCT06171802)	Randomized, double-blinded placebo-controlled, trial	Empagliflozin 10 mg daily initiated post-TAVR and continued for 180 d	Severe AS Undergoing TAVR	Change in LV mass index (cardiac CT) from baseline to 6 mo post-AVR	206	Recruitment ongoing
DAPAS (NCT05241431)	Randomized, double-blinded, placebo-controlled trial	Dapagliflozin 10 mg daily commenced within 1 mo before TAVR and continued for 13 mo	Severe AS undergoing TAVR LVEF 40–50%, or $\geq$ 50% with at least one of $\downarrow$ LV GLS, LVH or $\uparrow$ NT-proBNP	Composite of 10% reduction in LV mass index, 2.0% reduction in LV GLS, 25% reduction in NT-proBNP, and 10% improved in eGFR at 1 y post-TAVR	NR	Recruitment ongoing
SGLT2 inhibitors in moderate AS (NCT06469645)	Randomized open-label blinded end point controlled trial	Empagliflozin 10 mg daily for 6 mo	Moderate AS	Change in indexed extracellular volume from baseline to 6 mo	104	Recruitment ongoing
REVERSE-TAVR (NCT07090343)	Phase III, randomized, double-blind, placebo-controlled	Semaglutide 2.4 mg weekly initiated 3-months before TAVR and continuing for 24 mo post	Severe AS Undergoing TAVR BMI $\geq$ 30 kg/m ² or $\geq$ 27 kg/m ² and one of: dysglycemia, hypertension, hypercholesterolemia, OSA, history of stroke, history of myocardial infarction, peripheral arterial disease	Composite of cardiovascular death, nonfatal myocardial infarction, nonfatal stroke or transient ischemic attack, and heart failure hospitalization at 1 y and 27 mo	826	Study registered, recruitment not commenced

AS indicates aortic stenosis; AVR, aortic valve replacement; BMI, body mass index; CT, computed tomography; eGFR, estimated glomerular filtration rate; GLS, global longitudinal strain; HR, hazard ratio; LAE, left atrial enlargement; LV, left ventricular; LVEF, left ventricular ejection fraction; LVH, left ventricular hypertrophy; NCT, National Clinical Trial; NR, not reported; NT-pro-BNP, N-terminal pro-B-type natriuretic peptide; OSA, obstructive sleep apnea; SGLT2, sodium-glucose cotransporter 2; and TAVR, transcatheter aortic valve replacement.

CONCLUSIONS

AS and HFpEF frequently coexist, sharing risk factors, mechanisms, and similar patterns of myocardial dysfunction. Though historically regarded as distinct conditions, a growing body of evidence suggests that they interact along a pathophysiological continuum, producing overlapping phenotypes with important implications for diagnosis, treatment, and prognosis. This intersection is particularly relevant in older, comorbid populations, in whom both conditions are prevalent. Persistent HF after AVR remains common and challenging and, in many cases, may reflect coexisting and unrecognized HFpEF. In such cases, AVR alone may be insufficient, and therapies proven effective in HFpEF may offer benefit. Moving forward, improved phenotyping and risk stratification will be essential to delineate the relative contributions of AS and HFpEF in individual patients. Ultimately, an integrated approach that accounts for valvular, myocardial, and systemic contributors will be key to improving

outcomes in this increasingly prevalent and complex patient group.

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Disclosures

None.

Supplemental Material

Tables S1–S2

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